

<page_start no = 1>

<sol_start id=1>

1. Parallel beam of light

<sol_end>

<sol_start id=2>

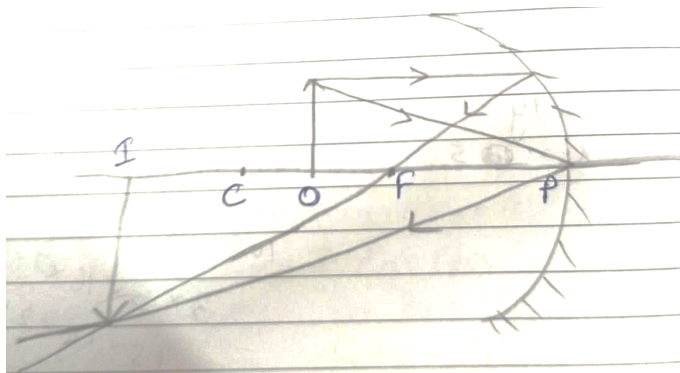
2. $f = \frac{R}{2}$

$\Rightarrow f = 25 \text{ cm}$

It will provide convex mirror as outward bulging surface of the sphere is to be used as mirror.

<sol_end>

<sol_start id=3>



3.

When the object is placed between C and f the image will be formed beyond C and would real, enlarged and inverted.

<sol_end>

<page_end>

<page_start no = 2>

<sol_start id=4>

4. height of object = 2 cm

$f = 10 \text{ cm}$

$u = 15 \text{ cm}$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$= \frac{1}{10} = \frac{1}{15} + \frac{1}{v}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{10} - \frac{1}{15}$$

$$\Rightarrow \frac{1}{v} = \frac{3-2}{30} = \frac{1}{30}$$

$$\therefore v = 30 \text{ cm}$$

Image is formed at a distance of 30 cm from the mirror.

$$m = \frac{-v}{u} = \frac{-30}{15} = -2$$

The image of the is diminished and it real and inverted.

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